MA 324 | END-SEMESTER EXAMINATION

STATISTICAL INFERENCE AND MULTIVARIATE ANALYSIS | 14:00–17:00 IST

IIT GUWAHATI | MAY 8, 2026

PART-A

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

FOR OFFICE USE ONLY

Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	6	9	20

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$\begin{aligned} Y_1 &= \theta_1 + \theta_2 + \epsilon_1, \\ Y_2 &= \theta_1 - 2\theta_2 + \epsilon_2, \\ Y_3 &= 2\theta_1 - \theta_2 + \epsilon_3, \end{aligned}$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2.

Ans:

~~x_i | y_i~~

$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad , \quad i = 1, 2, 3$

Objective: $\sum_{i=1}^{3} \varepsilon_i^2 = \sum_{i=1}^{3} (Y_i - \beta_0 - \beta_1 x_i)^2$

Objective: $\sum_{i=1}^{3} \varepsilon_i^2 = (Y_1 - \theta_1 - \theta_2)^2 + (Y_2 - \theta_1 + 2\theta_2)^2 + (Y_3 - 2\theta_1 + \theta_2)^2$

Our Job is to minimize it (loss function) to find LSE of θ_1 and θ_2.

(1) Differentiating wrt θ_1

$$-2(Y_1 - \theta_1 - \theta_2) - 2(Y_2 - \theta_1 + 2\theta_2) - 4(Y_3 - 2\theta_1 + \theta_2) = 0$$

$$\Rightarrow -2Y_1 + 2\theta_1 + 2\theta_2 - 2Y_2 + 2\theta_1 - 4\theta_2 - 4Y_3 + 8\theta_1 - 4\theta_2 = 0$$

$$\Rightarrow -2Y_1 - 2Y_2 - 4Y_3 + 12\theta_1 - 6\theta_2 = 0$$

$$\Rightarrow 12\theta_1 = 2Y_1 + 2Y_2 + 4Y_3 + 6\theta_2$$

$$\Rightarrow \boxed{\hat{\theta}_1 = \frac{1}{6}(Y_1 + Y_2 + 2Y_3 + 3\theta_2)} \quad \text{--- (i)}$$

(2)

$$L = (y_1 - \theta_1 - \theta_2)^2 + (y_2 - \theta_1 + 2\theta_2)^2 + (y_3 - 2\theta_1 + \theta_2)^2$$

Differentiating wrt θ_2:

$$-2(y_1 - \theta_1 - \theta_2) + 4(y_2 - \theta_1 + 2\theta_2) + 2(y_3 - 2\theta_1 + \theta_2) = 0$$

$$\Rightarrow -2y_1 + 2\theta_1 + 2\theta_2 + 4y_2 - 4\theta_1 + 8\theta_2 + 2y_3 - 4\theta_1 + 2\theta_2 = 0$$

$$\Rightarrow -2y_1 + 4y_2 + 2y_3 - 6\theta_1 + 12\theta_2 = 0$$

$$\Rightarrow 12\theta_2 = 2y_1 - 4y_2 - 2y_3 + 6\theta_1$$

$$\Rightarrow \boxed{\hat{\theta}_2 = \frac{1}{6}(y_1 - 2y_2 - y_3 + 3\theta_1)} \quad \text{--(ii)}$$

Putting (i) in (ii):

$$6\hat{\theta}_2 = y_1 - 2y_2 - y_3 + \frac{\cancel{3}}{\cancel{6}\,2}(y_1 + y_2 + 2y_3 + 3\theta_2)$$

$$12\theta_2 = 2y_1 - 4y_2 - \cancel{2y_3} + y_1 + y_2 + \cancel{2y_3} + 3\theta_2$$

$$9\theta_2 = 3y_1 - 3y_2$$

$$\boxed{\hat{\theta}_2 = \frac{1}{3}(y_1 - y_2)} \quad \text{--(iii)}$$

Putting (iii) in (i):

$$6\theta_1 = y_1 + \cancel{y_2} + 2y_3 + y_1 - \cancel{y_2}$$

$$\boxed{\hat{\theta}_1 = \frac{1}{3}(y_1 + y_3)}$$

$\therefore$ LSE's $\quad \hat{\theta}_1 = \frac{1}{3}(y_1 + y_3) \qquad \hat{\theta}_2 = \frac{1}{3}(y_1 - y_2)$

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2. Let U_1 and U_2 be two independent uniform random variables on the interval (0, 1). Suppose that $\boldsymbol{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.
   (a) (3 points) Compute the correlation matrix $\boldsymbol{\rho}$ of $\boldsymbol{X}$.
   (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

Ans-2. (a) $U \overset{i.i.d}{\sim} \text{Uniform}(\cancel{0,1})$

$\cancel{E[U] = 1/2}$ $\boxed{Var[U] = \frac{1}{12}}$

$Var(X_1) = \frac{1}{12} = Var(X_2)$

$$
\begin{aligned}
Var(X_3) &= Var(U_1 + U_2) \\
&= Var(U_1) + Var(U_2) + 2\underbrace{Cov(U_1, U_2)} \\
&= \frac{1}{12} + \frac{1}{12} + 0 \\
&= \frac{1}{6}
\end{aligned}
$$

~~cov~~ as independent random variables $\therefore Cov(U_1, U_2) = 0$

$$
\Sigma_X = \begin{bmatrix} 1/12 & 0 & \frac{1}{12} \\ 0 & 1/12 & 1/12 \\ 1/12 & 1/12 & 1/6 \end{bmatrix}
$$

$$
Cov(X_1, X_3) = Cov(U_1, U_1 + U_2) = Cov(U_1, U_1) + \cancel{Cov(U_1, U_2)}^{0}
$$
$$
= Var(U_1) = \frac{1}{12}
$$

$$
Cov(X_2, X_3) = Cov(U_2, U_1 + U_2) = Var(U_2) = 1/12.
$$

Now $\rho = \dfrac{Cov(x_i, x_j)}{\sqrt{Var(x_i)\,Var(x_j)}}$

$\rho_{x_1, x_3} = \dfrac{1/12}{\sqrt{\frac{1}{12} \cdot \frac{1}{6}}} = \dfrac{1}{\sqrt{2}}$

$\rho_{x_2, x_3} = \dfrac{1/12}{\sqrt{\frac{1}{12} \cdot \frac{1}{6}}} = \dfrac{1}{\sqrt{2}}$

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$$
\rho_X = \begin{bmatrix} 1 & 0 & 1/\sqrt{2} \\ 0 & 1 & 1/\sqrt{2} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix}
$$

(b)

$$\Sigma = \begin{bmatrix} 1 & 0 & 1/\sqrt{2} \\ 0 & 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1 \end{bmatrix}$$

first principal component $= (Y_1 = e_1' X)$

$\mathrm{Var}(Y_1) = e_1' \Sigma e_1 = \lambda_1$

$\because$ variance of first principal component is its largest eigenvalue (λ_1).

$|\Sigma - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 0 & 1/\sqrt{2} \\ 0 & 1-\lambda & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1-\lambda \end{vmatrix} = 0$$

$\Rightarrow (1-\lambda)\left((1-\lambda)^2 - \frac{1}{2}\right) + \frac{1}{\sqrt{2}}\left(\frac{\lambda - 1}{\sqrt{2}}\right) = 0$

$\Rightarrow (1-\lambda)^3 - \frac{(1-\lambda)}{2} - \frac{(1-\lambda)}{2} = 0$

$\Rightarrow (1-\lambda)^3 = (1-\lambda)$

$\Rightarrow (1-\lambda)\left[(1-\lambda)^2 - 1\right] = 0$

$\Rightarrow (1-\lambda) = 1$, $\quad (1-\lambda)^2 = 1$

$\boxed{\lambda = 0}$ $\quad 1 - \lambda = \pm 1$

$\lambda = 0, 2$

$\because$ Largest λ is 2

$\therefore$ $\boxed{\lambda_1 = 2}$ $\rightarrow$ variance of first principal component.

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3. Let the matrix of distance be

$$\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ 1 & 0 & & & \\ 2 & 1 & 0 & & \\ 3 & 11 & 2 & 0 & \\ 4 & 5 & 3 & 3.5 & 0 \end{array}$$

(a) (6 points) Cluster the four items using each of the following procedures:
   i. Complete linkage hierarchical procedure.
   ii. Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

Ans-3. (a)

$$\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ 1 & 0 & & & \\ 2 & 1 & 0 & & \\ 3 & 11 & 2 & 0 & \\ 4 & 5 & 3 & 3.5 & 0 \end{array}$$

minimum distance = 1 between 1 & 2

∴ (1 2) will form cluster in starting.

(i) Complete linkage

$d_{(UV)W} = \max(d_{UW}, d_{VW})$

↳ for complete linkage

$$\begin{array}{c|ccc} & (12) & 3 & 4 \\ (12) & 0 & & \\ 3 & 11 & 0 & \\ 4 & 5 & 3.5 & 0 \end{array}$$

Minimum distance = 3.5 between 3 & 4.

(3 4) will form cluster

↓

$$\begin{array}{c|cc} & (12) & (34) \\ (12) & 0 & \\ (34) & 11 & 0 \end{array}$$

for final cluster (1 2 3 4)

minimum distance = 11

(ii) Average Linkage

$$d_{(UV)W} = \frac{\sum \sum d_{UW} + \sum d_{VW}}{N_{UV} N_W}$$

$$\begin{array}{c c} & \begin{array}{ccc} (12) & 3 & 4 \end{array} \\ \begin{array}{c} (12) \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 & & \\ 6.5 & 0 & \\ 4 & 3.5 & 0 \end{bmatrix} \end{array}$$

Minimum distance = 3.5 to form cluster (3, 4).

↓

$$\begin{array}{c c} & \begin{array}{cc} (12) & (34) \end{array} \\ \begin{array}{c} (12) \\ (34) \end{array} & \begin{bmatrix} 0 & \\ 21/4 = 5.25 & 0 \end{bmatrix} \end{array}$$

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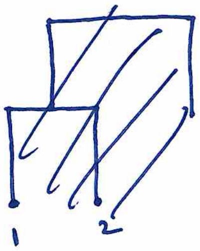
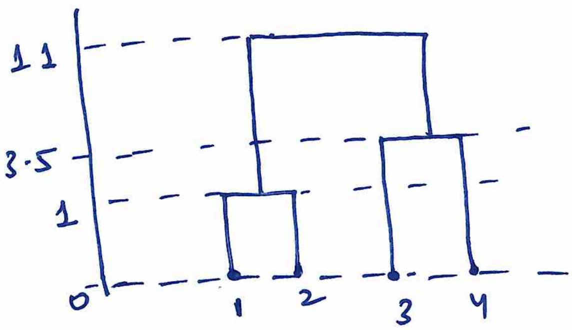
Minimum distance = $\frac{21}{4}$ = 5.25 to form (1, 2, 3, 4)

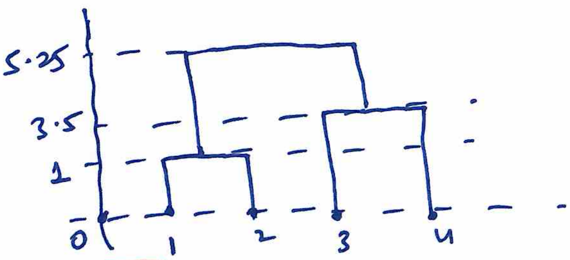
(b) Dendrogram i →





Complete linkage clustering.

Dendrogram ii →



03 Average linkage

Results Comparison

① On first two iterations, both clusters are formed same, i.e. (1 2) and (3 4) at distances 1 and 3.5 respectively.

② but final cluster is formed (1 2 3 4) at 11 distance in complete & distance 5.25 at in average linkage as complete is very tight & it needs max of distances to be taken.

Additional Pages for Answers